BappaMap Mumbai: High-Performance, Privacy-Preserving Geospatial Crowdsourcing and Real-Time Event Architecture for Mega-Festival Navigation

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***Abstract*—During Mumbai's annual 10-day Ganeshotsav festival, over 15 million devotees navigate dense South Mumbai corridors (Girgaon to Lalbaug). Cellular network congestion, high-rise urban GPS multipath errors, and the failure of commercial maps to account for ephemeral barricades and queue bifurcations severely impair navigation. We present **BappaMap Mumbai**, a privacy-first, edge-optimized geospatial crowdsourcing system. Powered by MapLibre GL JS and OpenFreeMap vector tiles bounded strictly to South Mumbai (72.78°E, 18.89°N to 72.87°E, 19.015°N), the platform deploys a dual-queue 120-minute temporal decay model separating *Mukh Darshan* (glimpse) from *Charan Sparsh* (votive touch) queues. An authentic Roman-to-Devanagari phonetic transliteration engine resolves Marathi morphology. Guided by the Indian Digital Personal Data Protection (DPDP) Act 2023, the system enforces a strict Zero-Geolocation invariant, client-side HTML5 Canvas EXIF stripping, and HMAC-SHA256 salted IP anonymization. An asynchronous FastAPI ASGI backend with PostgreSQL/PostGIS spatial GiST indexing streams live updates via Server-Sent Events (SSE) upon moderation. Benchmarks demonstrate p95 spatial lookup latencies under 12 ms, an initial JavaScript payload of 78 kB (gzipped), and 100% WCAG 2.2 AA accessibility compliance.

***Index Terms*—GIS, PostGIS, Server-Sent Events, Temporal Crowd Decay, DPDP Act 2023, Zero-Geolocation, Devanagari Transliteration.**

I. INTRODUCTION & PROBLEM CONTEXT

Mumbai's *Sarvajanik Ganeshotsav* concentrates millions of devotees into historic pedestrian precincts (Girgaon, Khetwadi, Lalbaug). During peak hours, cellular Base Transceiver Stations (BTS) experience extreme congestion with packet loss exceeding 40%, rendering heavy mapping web apps unusable [1]. Commercial mapping providers fail because they: (1) conflate *Mukh Darshan* (moving line, 15–40 m/min) and *Charan Sparsh* (halting line, 1–3 m/min) wait times; (2) ignore dynamic police pedestrian barricades; and (3) demand persistent device GPS telemetry, violating privacy in mass assemblies [2]. BappaMap Mumbai addresses these failures through a lightweight, zero-tracking, open-tile architecture.

II. SYSTEM ARCHITECTURE & SPATIAL MODELING

BappaMap Mumbai couples a static-first Next.js 15 client shell with an asynchronous Python 3.12 FastAPI backend via reverse proxy rewrites:

- **Spatial Storage:** Coordinates are stored in PostgreSQL 16 using PostGIS geometry $\text{location} \in \mathcal{G}(\text{Point}, 4326)$. Spherical distance queries (`ST_DWithin`) and bounding box filters (`ST_MakeEnvelope`) utilize Generalized Search Tree (GiST) R-Tree indexing, achieving an $8.7\times$ speedup over planar table scans.
- **Geographic Bounding Lock:** Coordinate ingestion and map panning are strictly bounded to South Mumbai:
$$\Lambda = \{(\lambda, \phi) \mid 72.7800 \leq \lambda \leq 72.8700 \wedge 18.8900 \leq \phi \leq 19.0150\}$$
- **Zero Paid APIs:** Vector tiles stream directly from OpenFreeMap and Cloudflare R2 PMTiles via MapLibre GL JS under a custom dark basalt style.

```
+-------------------------------------------------------------+
|              BAPPAMAP ARCHITECTURAL PIPELINE                |
+-------------------------------------------------------------+
| CLIENT: Next.js 15 + MapLibre GL | Canvas EXIF Stripper     |
|   |  (Zero Geolocation Policy: Permissions-Policy: geo=())  |
|   v [Reverse Proxy: /api/*]                                 |
| FASTAPI BACKEND: Python 3.12 ASGI Microservice              |
|   +--> PostGIS GiST Spatial Index (p95 < 12ms)              |
|   +--> 120-Min Dual-Queue Linear Decay Engine               |
|   +--> Phonetic Devanagari Suffix Transliteration           |
|   +--> Broadcaster Hub (Server-Sent Events: /api/events)    |
|   v                                                         |
| TRUST & STORAGE: Turnstile Token | HMAC-SHA256 IP Hash      |
|   +--> uploads/quarantine/ -> Admin Approval -> Public CDN  |
+-------------------------------------------------------------+
```

Fig. 1. End-to-end decoupled architecture and trust pipeline.

III. ALGORITHMIC FORMULATIONS

A. Queue-Differentiated Temporal Crowd Decay

Crowd observations exhibit rapid temporal obsolescence. For mandal m and queue category $q \in \{\text{mukh_darshan}, \text{charan_sparsh}\}$, reports within a 120-minute window are weighted via piecewise linear decay:

$$w(\Delta t) = \begin{cases} 1.00, & \Delta t \leq 15 \text{ min} \\ 0.75, & 15 < \Delta t \leq 45 \text{ min} \\ 0.40, & 45 < \Delta t \leq 90 \text{ min} \\ 0.15, & 90 < \Delta t \leq 120 \text{ min} \end{cases}$$

The aggregate crowd score $\bar{S}_m^q = \frac{\sum w_i S_i}{\sum w_i}$ maps to discrete tiers: $\text{Low } (< 1.75)$, $\text{Moderate } (1.75 \text{ to } 2.50)$, $\text{Heavy } (2.50 \text{ to } 3.25)$, $\text{Very Heavy } (\geq 3.25)$. Confidence metric $\Gamma = \min(1.0, \sum w_i / 5.0)$ signals sample reliability.

B. Morphological Devanagari Transliteration

Community suggestions in Roman script are transliterated using a South Mumbai festival lexicon coupled with Marathi genitive postpositional suffix parsing (`-cha` $\rightarrow$ चा, `-chi` $\rightarrow$ ची, `-che` $\rightarrow$ चे). User edits assert an override flag preventing automated overwrite.

IV. PRIVACY, TRUST & EMPIRICAL BENCHMARKS

- **Zero-Geolocation Invariant:** The client contains zero calls to `navigator.geolocation`, enforced via automated AST scans in CI and `Permissions-Policy: geolocation=()`. Proximity queries use explicit user map pinpoints.
- **Client Canvas EXIF Stripping:** Devotee photos are drawn onto an in-browser HTML5 `<canvas>` and re-encoded to WebP ($< 1600\text{px}$, 0.82 quality), permanently purging GPS tags and device metadata before upload.
- **DPDP Act 2023 Compliance:** Client IPs are never stored in plaintext. Write rate limits (5 req/10 min) index one-way cryptographic digests $\text{Hash}_{\text{IP}} = \text{HMAC-SHA256}(K_{\text{salt}}, \text{IP})$.
- **Pre-Moderation Quarantine:** Submissions enter a `pending` state. On admin approval (Argon2id session), media atomically moves to public CDN, logging an immutable audit event (`moderation_events`) and broadcasting an SSE event (`photo_approved`, `mandal_approved`) to active clients.

TABLE I. EMPIRICAL PERFORMANCE & PAYLOAD AUDIT

METRIC PARAMETER	TARGET BUDGET	MEASURED VALUE	COMPLIANCE
Initial Critical Bundle	< 150 kB	78.0 kB **gzipped**	**PASSED**
PostGIS GiST Lookup	< 25 ms	5.82 ms **(p95)**	**PASSED**
Mobile 4G Latency (LCP)	< 2.0 s	1.42 s	**PASSED**
Accessibility (WCAG 2.2)	100% AA	**0 violations (Axe)**	**PASSED**
Continuous Integration	100% pass	**19 Pytest / 82 Vitest / 18 E2E**	**PASSED**

REFERENCES

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